



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

June 20, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 12098
(FOURTEENTH REVISION)

EXPIRATION DATE: 2026-01-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Mission Systems Orchard Park Inc.
Orchard Park, NY
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale, and use of a non-DOT specification cylinder conforming with all regulations applicable to a DOT Specification 3HT cylinder for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing special permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.301(f) in that each non-DOT specification cylinder is not equipped with a pressure relief device and § 173.302a(a)(1) in that the prescribed non-DOT specification cylinder is not an authorized packaging, except as specified herein.

5. BASIS: This special permit is based on the application of Mission Systems Orchard Park Inc. dated December 4, 2024, submitted in accordance with § 107.105 and the public proceeding thereon.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Oxygen, compressed	2.2	UN1072	N/A
Compressed gas, oxidizing, n.o.s.	2.2	UN3156	Hazard Zone D
Helium compressed	2.2	UN1046	N/A
Nitrogen, compressed	2.2	UN1066	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Prescribed packaging is a non-DOT specification welded sphere that must be transported installed in a re-breather apparatus. The sphere must conform with Carleton Technologies Inc.'s drawing B16313-3 dated December 19, 2018, on file with the Office of Hazardous Materials Safety (OHMS), and must be fabricated in accordance with the DOT-3HT Specification (§§ 178.35 and 178.44), except as follows:

§ 178.35(e) *Safety devices*. The individual sphere is not equipped with a pressure relief device; the re-breather apparatus is equipped with a pressure relief device that limits the pressure in the sphere to 4,375 psig or less.

§ 178.35(f) *Markings*.

(1) * * *

(i) Each cylinder must be marked "DOT-SP 12098" in lieu of "DOT-3HT", followed by the service pressure.

§ 178.44(a) *Type, size, and service pressure*. A welded sphere with a maximum volume of 180 cubic inches and a maximum service pressure of 3,500 psig.

§ 178.44(b) *Authorized steel.* The cylinder must be constructed of Inconel 718 in accordance with UNS N07718.

§ 178.44(e) *Welding.*

- (1) Qualification of welding operations & weldments.
 - (i) Welding of the two spherical halves is authorized. Welding procedures and operations must be qualified in accordance with SAE-AMS-STD-2219, Class A.
 - (ii) Welding of the neck fitting to the top spherical half is authorized. Welding must be in accordance with AMS 2681.
- (2) Radiography of weldments.
 - (i) Welding of spherical halves must be 100% radiographically examined in accordance with ASTM-E-1742. Interpretation and acceptance of radiographs must be in accordance with MIL-HDBK-1264, Grade II.
 - (ii) Welding of neck fitting (port fitting) to the top spherical half must be 100% radiographically examined in accordance with ASTM-E 1742. The interpretation and acceptance of radiographs must be in accordance with AWS D17.1, Class A.
- (3) All welded joints must be liquid penetrant inspected in accordance with ASTM-E-1417, Type 1, Method A, B, or C, Sensitivity Level 1. Acceptance of liquid penetrant results must be in accordance with MIL-STD-1907, Grade A.

§ 178.44(f) *Wall thickness.*

- (1) The minimum wall thickness must be such that the wall stress at minimum specified test pressure may not exceed 135,000 psi.
- (2) Calculations must be made by the formula:

$$S = PD/4tE$$

Where:

S = Wall stress in psi;

P = Test pressure prescribed for water jacket test, in psi;

D = Outside diameter in inches;

t = Minimum wall thickness in inches;

E = 0.85 (provides 85 percent weld efficiency factor which must be applied in the girth weld area and heat affected zones. The heat affected zone must extend a distance of 6 times wall thickness from center of weld).

(3) Does not apply.

§ 178.44(g) *Heat treatment*. The completed cylinders must be uniformly and properly heated prior to tests.

§ 178.44(j) *Cycling test*.

(1) * * *

(2) * * *

(3) A lot is defined as a group of cylinders fabricated from the same material, manufactured by the same process and heat treated in the same equipment under the same conditions of time, temperature, and atmosphere, and may not exceed a quantity of 200 cylinders.

(4) * * *

§ 178.44(l) *Flattening test*.

(1) One sample ring representative of the cylinder must accompany each lot. The sample ring must be from the same material and be subjected to the same heat treatment as the lot of cylinders.

(2) The sample ring representative of the cylinder must be flattened. The sample ring may not include the heat affected zone or any weld.

(3) Flattening between flat plates is authorized.

(4) If the sample ring cracks when subjected to specified flattening test, the lot of cylinders represented by the test must be rejected (see § 178.44(q)).

§ 178.44(m) *Physical tests*. Physical tests are not required.

§ 178.44(n) *Magnetic particle inspection*. Not required. Instead, each cylinder must be inspected using apparatus and procedures for liquid penetrant examination in accordance with ASTM E-165. Inspection must be performed externally on the finished cylinder after hydrostatic testing. Evidence of discontinuities, which in the opinion of the independent inspector may appreciably weaken or decrease the durability of the cylinder, must be cause for rejection.

§ 178.44(o) *Leakage test*. Does not apply.

§ 178.44(p) *Acceptable results of tests*.

- (1) Flattening: The sample ring may not crack when flattened so that outer surfaces are not more than fifteen times wall thickness apart.
- (2) Physical tests: N/A
- (3) The burst pressure must be at least 1.38 times the test pressure.
- (4) * * *

§ 178.44(r) *Marking*.

- (1) Cylinders must be marked by low stress type steel stamping or laser engraving in an area and to a depth which will insure that the wall thickness measured from the root to the interior surface is equal to or greater than the minimum prescribed wall thickness. Marking must be permanent and legible. Marking on side wall not authorized.

b. TESTING: Each cylinder must be requalified in accordance with the requirements for DOT 3HT cylinders as specified in § 180.209(g).

c. OPERATIONAL CONTROLS:

- (1) A copy of the inspector's report required by § 178.35(g) for the first lot of cylinders must be submitted to the OHMS prior to first shipment.
- (2) Cylinders must be shipped in strong outside packagings in accordance with § 173.301(a)(9).

(3) Each cylinder is authorized for use under this special permit for no more than 24 years from the date of manufacture. Cylinders must be removed from service at the end of the authorized life.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the OHMS for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each facility where the packaging is manufactured under this special permit. It must be made available to a DOT representative upon request.

f. The transportation of oxygen by aircraft is only authorized when in accordance with § 175.501.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, and cargo-only aircraft.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

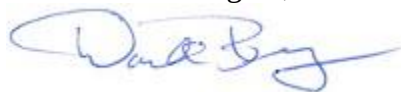
Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Quade
Acting Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: TG/BenM